

ZONIRAZ JEWELHOUSE PRIVATE LIMITED

A SYMPHONY OF BRILLIANCE AND ELEGANCE

Document Type: Full Project Technical & System Specification

Company Name: Zoniraz Jewelhouse Private Limited

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GSTIN: 08AABCZ4653J1ZY | **State Code:** 08 | **PAN:** AABCZ4653J

Core Subsystems: Frontend (React 19), Backend (Node.js/Express), Admin Panel (TypeScript)

Total Source Code Files: 248 Source Code Files (47,764 Files Including Node Modules)

Primary Typography: Playfair Display (Serif Titles) & Inter (Sans-Serif UI Body)

Executive Summary:

This comprehensive document contains the complete technical architecture, subsystem file breakdown, database schemas, API service logic, pricing algorithms, Gold Savings Scheme (Gold Mine 10+1) mechanics, email invoice generation rules, WebRTC live video consultation flow, and UI typography standards for the **Zoniraz Jewels** e-commerce platform.

1. System Subsystems & Directory Statistics

The platform is structured into three primary decoupled subsystems: Customer Frontend, Node.js Backend Server, and TypeScript Admin Panel.

Subsystem	Primary Tech Stack	Source Files	Total Files	Core Function & Responsibility
Customer Frontend	React 19, Vite, JS/JSX, Vanilla CSS, Lucide Icons	128 Files	9,687 Files	Customer store, custom rate/weight calculator, Gold Mine portal, Razorpay checkout, digital gold wallet
Backend Server	Node.js, Express, MongoDB, Nodemailer, Socket.io	95 Files	31,460 Files	REST API backend, live pricing calculation engine, Gold Mine scheme, PDF/HTML invoice generator with embedded Base64 badges, WebRTC signaling
Admin Panel	React 19, TypeScript, Vite, TailwindCSS 4	25 Files	6,617 Files	Admin management portal, masterpiece catalog editor, orders ledger, 10+1 wallets control, live pricing editor
TOTAL PROJECT	MERN Stack + WebRTC + Nodemailer PDF	248 Source Files	47,764 Total	Full Luxury Jewellery E-Commerce Ecosystem

2. Typography, Fonts & Design Tokens

Typography Standards:

- **Primary Header Font (Serif):** Playfair Display (Google Fonts - Weights 400..700, Italic) — Renders luxury product titles, collection headings, section banners, price tags, and logo branding.
- **Body & UI Font (Sans-Serif):** Inter (Google Fonts - Weights 300, 400, 500, 600, 700) — Renders navigation drawers, product specification tables, buttons, form controls, order receipts, and admin tables.

Design Color Palette:

- **Deep Luxury Cocoa Brown:** #5d463c / #231535 (Primary brand accent for headers, primary buttons, and sidebar navigation)
- **Royal Gold:** #c8a359 / #C5A880 / #B7791F (Secondary highlight for prices, Gold Mine wallet balances, and badges)
- **Cream Linen & Warm Canvas:** #efe7e5 / #FAF5EF / #f4f6f9 (Background for pages, modals, and product cards)
- **Favicon Icon:** /zoni1.png — Official Zoniraz brand logo favicon configured in frontend/index.html and adminSide/index.html.

3. Realtime Dynamic Pricing & Weight Calculation Engine

The backend service `backend/src/services/productPriceCalculation.js` contains the custom formula that computes live jewellery prices and dynamic net/gross weights whenever a customer changes Ring/Chain Size or Karat Purity on `ProductDetailPage.jsx`.

Formula & Rules:

- 1. Normalized Input Handling:** Metal selections (e.g. 9K, 14K, 18K, 22K, 24K) and Diamond Grades (IJ-SI, GH-VS, EF-VVS, FG-SI) are normalized to prevent default fallback errors.
- 2. Size Weight Adjustment:** For Rings, every size step past default Size 12 adds +0.140g gold weight. For Chains/Mangalsutras, every 1 inch past base length adds +0.500g.
- 3. Karat Density Multipliers:**
 - 9K Gold = 37% 24K Rate | Weight Multiplier = $37 / 58.5$
 - 14K Gold = 58.5% 24K Rate | Weight Multiplier = 1.00 (Base)
 - 18K Gold = 75% 24K Rate | Weight Multiplier = $75 / 58.5$
 - 22K Gold = 91.6% 24K Rate | Weight Multiplier = $91.6 / 58.5$
 - 24K Gold = 100% 24K Rate | Weight Multiplier = $100 / 58.5$
- 4. Net Gold Weight Calculation:**
$$\text{Net Gold Weight} = \text{Gross Gold Weight} - (\text{Diamond Weight ct} * 0.2) - (\text{Solitaire Weight ct} * 0.2) - (\text{Gemstone Weight ct} * 0.2)$$
- 5. Total Price Assembly:**
$$\text{Total Base Price} = (\text{Net Gold Weight} * \text{Gold Rate}) + (\text{Diamond Weight} * \text{Diamond Rate}) + \text{Solitaire Price} + \text{Gemstone Price} + \text{Making Charges}$$

$$\text{Final Customer Price} = \text{Total Base Price} + 3\% \text{ GST}$$

4. Zoniraz 10+1 Gold Savings Scheme & Email Invoices

Zoniraz 10+1 Gold Mine Scheme:

- Customer deposits 10 consecutive monthly EMI installments.
- Each installment is automatically converted into 24K pure gold grams based on the live gold rate on payment date and stored in customer's UserDashboard Gold Wallet.
- **100% Free 11th Bonus Month:** If all 10 EMIs are paid on time (within 30-day grace period), Zoniraz contributes the 11th monthly installment completely free.

Email Invoice & Base64 Certification Badges (`goldMineEmailService.js`):

- Invoices & statements are sent via Nodemailer to customer and admin.
- The footer section embeds 9 Base64-encoded Trust & Certification Badges: Assured Purity, 22K916 Hallmark, SGL Certified, IGI Certified, Trust & Safety, Certified Quality, 100% Natural Diamond, BIS Hallmarked Logo, and Authentic Jewellery Guarantee.

5. Admin Portal & Live WebRTC Video Consultation

Admin Portal (adminSide/):

- **Dashboard:** Revenue counters, total active Gold Mine wallets, recent orders.
- **Catalog & Masterpiece Editor:** Add/edit products, stock levels, custom prices.
- **Patron Ledger (Customers.tsx):** Null-safe customer search, lifetime spend tracking, account status suspension/activation.
- **Gold Mine Wallets Manager:** View customer passbooks, credit manual EMIs, toggle 11th bonus status.
- **Daily Pricing Settings:** Update live 24K gold rate, diamond grade modifiers, GST rate.

WebRTC Live Video Consultation (VideoCallContext.jsx / VideoCallPanel.tsx):

- Uses Socket.io signaling server to connect customer browser with admin for live virtual jewellery try-on calls.